

Finding the equation of a parabola

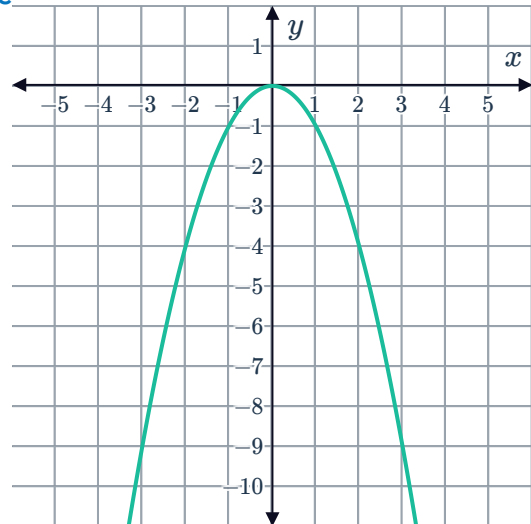


1

a $a = -1$

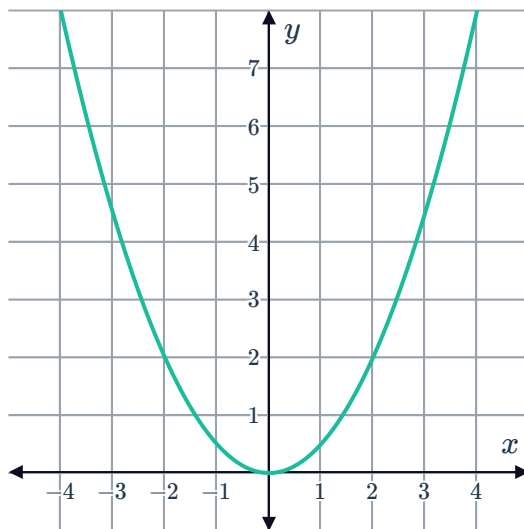
b $(0,0)$

c



2

a



b $y = \frac{1}{2}x^2$

3

a No

b Yes

4

a $(3,5)$

b $y = (x - 3)^2 + 5$

5

a $x = -7$

b $y = -\frac{3}{7}(x - 1)(x + 7)$

c $\left(-3, \frac{48}{7}\right)$



7

a $x = -2$

b $y = (x + 2)^2 - 9$

8 $y = x^2 + 6x - 40$

9

a $x^2 - b^2$

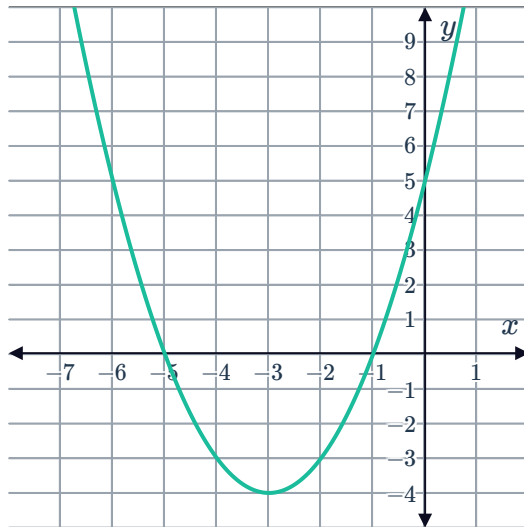
b $a = 0$

10

a $y = (x + 1)(x + 5)$

b $y = 5$

c

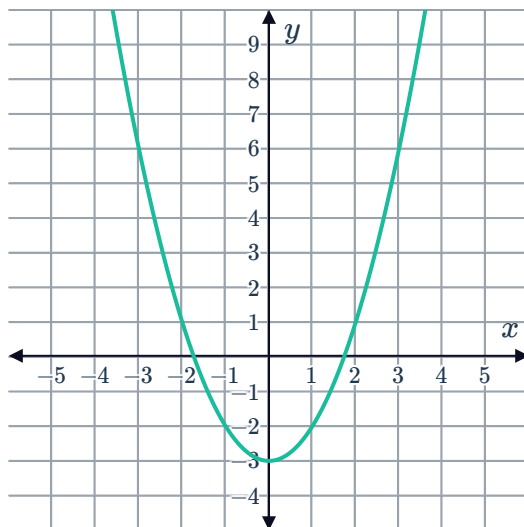


11

a $y = x^2 - 3$

b -2

c



12

a infinitely many

c 1



b infinitely many

$$d \quad y = -\frac{3x^2}{25} - \frac{6x}{25} + \frac{72}{25}$$

$$13 \quad y = \frac{x^2}{4} + 6x + 39$$

$$14 \quad b = 4$$

$$15 \quad c = 4$$

$$16 \quad k = \frac{1}{3}$$

17

$$a \quad a + b + c = 3, 9a + 3b + c = -1, 16a + 4b + c = 0$$

$$b \quad c = 8$$

$$c \quad a = 1$$

$$d \quad b = -6$$

$$e \quad x^2 - 6x + 8$$

Maximum and minimum values

18

a Maximum

$$b \quad (0, -4)$$

19

a Concave up

$$b \quad x = 9$$

$$c \quad y = -71$$

$$20 \quad y = 0$$

21

$$a \quad -0.087$$

$$b \quad -1.24$$

22

$$a \quad x = -6$$

$$b \quad m = 330$$

23

$$a \quad x = 10$$

$$b \quad m = 607$$



Graphing quadratics using technology

24 $y = (x - 2)^2 + 3$

25

a Concave up

b $x = 1, -5$

c $y = -5$

d $x = -2$

e $(-2, -9)$

26 $-\frac{81}{4}$

27

a $x = -3.1$

b -16.61

28

a Concave up

b $y = 9$

c $x = -3$

d $x < -3$

29

a Concave down

b $y = -6$

c $x = -3$

d $x < -3$

30

a $(-6, -9)$

b $(0, 135)$

Applications

31

a Negative a , positive c

b $c = 2560$

c $a = -\frac{1}{10}$

d $\frac{25\,071}{10}$ m

32

a $y = -\frac{1}{20}(x - 18)^2 + 16.2$

b 4071.50 m^2

33

a $A(0, 2.5), B(2, 0), C(6, 0)$

b 2

c $4a + k = \frac{5}{2}$

d $0 = 16a + k$

e 5

f 10



34

a $A(0, 3.5), B(3, 0), C(7, 0)$

c $9a + k = \frac{7}{2}$

e $a = -\frac{1}{2}$

b 3

d $0 = 16a + k$

f $k = 8$

35

a $(180, 98)$

c Below

b $y = -\frac{49}{16200}(x - 180)^2 + 98$

36

a $A(18, 0), B(0, 108)$

c $x = 9\sqrt{2}$

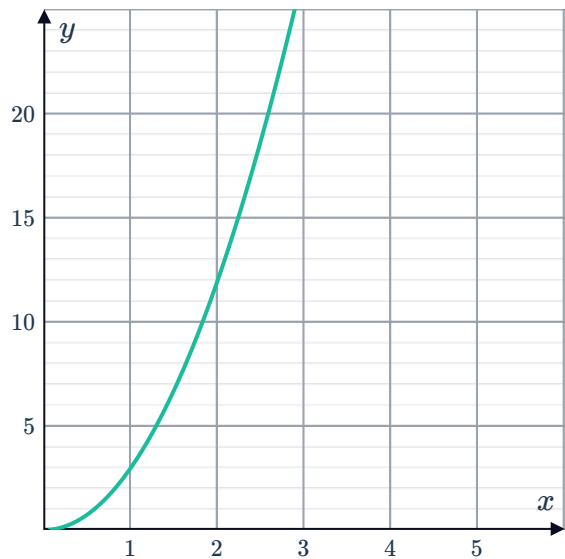
b $y = -\frac{1}{3}(x - 18)(x + 18)$

37

a

x	0	1	2	3	4
y	0	3	12	27	48

b



c 0 metres

e $(0, 0)$

d 0

38

a Initially, the object has not fallen any distance and is 700 m above the ground.

b decreasing

c

i True

ii True

iii False

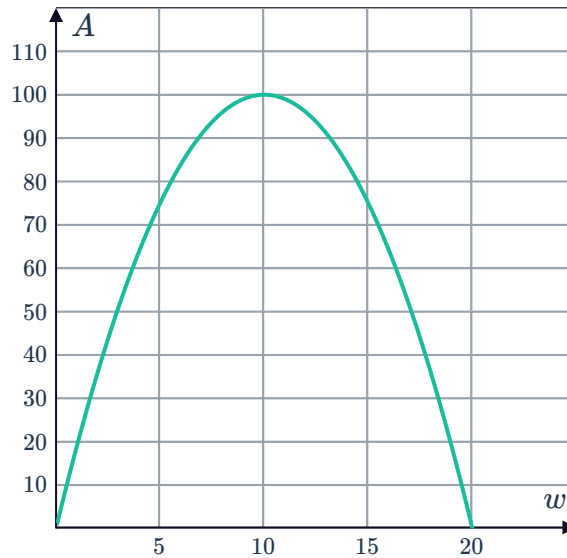
39



- a After 4.5 seconds, the pebble is 40.5 metres above the ground. The highest distance the pebble reaches above the ground is 40.5 metres.
- b $t = 0, 9$
- c The times when the pebble is on the ground.

40

a



- b $w = 0, 20$
- c There is no corresponding rectangle because its area would be 0.
- d 100

41

- a 0, 40
- b 20
- c The largest rectangular area that can be fenced off with 80 m of wire.

42

- a $s = 12$
- b 337

43

- a $t = 20$
- b 100 metres

44

- a $13 - x$
- b $x(13 - x)$
- c 6.5
- d 42.25 m^2



Applications using graphing calculators

45

a (0, 0)

b The initial distance fallen by the object.

46 $-14, 14 \text{ m/s}$

47

a 176.4 m

b 6 s

c 12 s

48

a 900 m

b Initially, the object has not fallen any distance.

c 13.55 s

d After 13.55 seconds the object is on the ground.

49

a 0, 30

b 15

c The largest rectangular area that can be fenced off with 60 m of wire.

50

a 4 s

b 10 s